

```
student@L08-C27:~/Desktop/2520071017$ mkdir os_lab
student@L08-C27:~/Desktop/2520071017$ cd os_lab
student@L08-C27:~/Desktop/2520071017/os_lab$ ls
student@L08-C27:~/Desktop/2520071017/os_lab$ cd ..
student@L08-C27:~/Desktop/2520071017$ cd os_lab
student@L08-C27:~/Desktop/2520071017/os_lab$ mkdir year1 year2 year3
student@L08-C27:~/Desktop/2520071017/os_lab$ ls
year1  year2  year3
student@L08-C27:~/Desktop/2520071017/os_lab$ rmdir year3
student@L08-C27:~/Desktop/2520071017/os_lab$ touch year1/file.txt
student@L08-C27:~/Desktop/2520071017/os_lab$ rmdir year1
rmdir: failed to remove 'year1': Directory not empty
student@L08-C27:~/Desktop/2520071017/os_lab$ rm year1/file1.txt
rm: cannot remove 'year1/file1.txt': No such file or directory
student@L08-C27:~/Desktop/2520071017/os_lab$ rm year1/file.txt
student@L08-C27:~/Desktop/2520071017/os_lab$ rmdir year1
student@L08-C27:~/Desktop/2520071017/os_lab$ echo "YourName RoINumber" > student.txt
student@L08-C27:~/Desktop/2520071017/os_lab$ cat student.txt
YourName RoINumber
student@L08-C27:~/Desktop/2520071017/os_lab$ echo -e "Mathematics\noperating Systems\nData Structures" > subject.txt
student@L08-C27:~/Desktop/2520071017/os_lab$ cat subject.txt
Mathematics
operating Systems
Data Structures
student@L08-C27:~/Desktop/2520071017/os_lab$ ls -l student.txt
-rw-rw-r-- 1 student student 19 Apr 10 12:02 student.txt
student@L08-C27:~/Desktop/2520071017/os_lab$ chmod 740 student.txt
student@L08-C27:~/Desktop/2520071017/os_lab$ ls -l student.txt
-rwxr----- 1 student student 19 Apr 10 12:02 student.txt
student@L08-C27:~/Desktop/2520071017/os_lab$ chmod 444 student.txt
student@L08-C27:~/Desktop/2520071017/os_lab$ echo "New Line" >> student.txt
bash: student.txt: Permission denied
student@L08-C27:~/Desktop/2520071017/os_lab$
```

```
student@L6015-C18:~/Desktop/2520071017$ rm mydir3/world.txt
student@L6015-C18:~/Desktop/2520071017$ rm -r mydir3
student@L6015-C18:~/Desktop/2520071017$ rm -r mydir2
student@L6015-C18:~/Desktop/2520071017$ mv mydir1/hello.doc .
student@L6015-C18:~/Desktop/2520071017$ rm -r mydir1
student@L6015-C18:~/Desktop/2520071017$ echo "Hello $USER." > greeting.doc
student@L6015-C18:~/Desktop/2520071017$ cat hello.doc greeting.doc > concat.out
student@L6015-C18:~/Desktop/2520071017$ cat concat.out
```

Hello World!

Hello student.

```
student@L6015-C18:~/Desktop/2520071017$ mv concat.out concat.doc
```

```
student@L6015-C18:~/Desktop/2520071017$ ls -l *.doc
```

```
-rw-rw-r-- 1 student student 28 Feb 19 11:42 concat.doc
```

```
-rw-rw-r-- 1 student student 15 Feb 19 11:41 greeting.doc
```

```
-rw-rw-r-- 1 student student 13 Feb 19 11:29 hello.doc
```

```
student@L6015-C18:~/Desktop/2520071017$
```

```
student@L6015-C18:~/Desktop/2520071017$ pwd
/home/student/Desktop/2520071017
student@L6015-C18:~/Desktop/2520071017$ echo $HOME
/home/student
student@L6015-C18:~/Desktop/2520071017$ ls
student@L6015-C18:~/Desktop/2520071017$ ls -l
student@L6015-C18:~/Desktop/2520071017$ mkdir mydir1
student@L6015-C18:~/Desktop/2520071017$ ls mydir1
student@L6015-C18:~/Desktop/2520071017$ cd mydir1
echo "Hello World!" > hello.txt
student@L6015-C18:~/Desktop/2520071017/mydir1$ cat hello.txt
Hello World!
student@L6015-C18:~/Desktop/2520071017/mydir1$ cp hello.txt hello.doc
student@L6015-C18:~/Desktop/2520071017/mydir1$ mkdir ../mydir2
mkdir ../mydir3
student@L6015-C18:~/Desktop/2520071017/mydir1$ cd ../mydir2
ls ..
mydir1 mydir2 mydir3
student@L6015-C18:~/Desktop/2520071017/mydir2$
student@L6015-C18:~/Desktop/2520071017/mydir2$ cp ../mydir1/hello.txt .
student@L6015-C18:~/Desktop/2520071017/mydir2$ mv hello.txt world.txt
student@L6015-C18:~/Desktop/2520071017/mydir2$ mv world.txt ../mydir3
student@L6015-C18:~/Desktop/2520071017/mydir2$ chmod a+w ../mydir3/world.txt
student@L6015-C18:~/Desktop/2520071017/mydir2$ cd ../mydir3
student@L6015-C18:~/Desktop/2520071017/mydir3$ chmod u-x ../mydir2
student@L6015-C18:~/Desktop/2520071017/mydir3$ rm ../mydir1/*.TXT
rm: cannot remove '../mydir1/*.TXT': No such file or directory
student@L6015-C18:~/Desktop/2520071017/mydir3$ rm ../mydir1/*.txt
student@L6015-C18:~/Desktop/2520071017/mydir3$ cd ~
```

C BankerAlgorithm.c X

C BankerAlgorithm.c > main()

```

48
49     for(i = 0; i < n; i++) {
50         if(finish[i] == 0) {
51             int flag = 0;
52
53             for(j = 0; j < m; j++) {
54                 if(need[i][j] > avail[j]) {
55                     flag = 1;
56                     break;
57                 }
58             }
59
60             if(flag == 0) {
61                 for(k = 0; k < m; k++) {
62                     avail[k] += alloc[i][k];
63                 }
64
65                 safeSeq[count++] = i;
66                 finish[i] = 1;
67                 found = 1;
68             }
69         }
70     }
71
72     if(found == 0) {
73         printf("\nSystem is NOT in safe state (Deadlock possible)\n");
74         return 0;
75     }
76 }
77
78 printf("\nSystem is in SAFE state\nSafe sequence is:\n");
79 for(i = 0; i < n; i++) {
80     printf("P%d ", safeSeq[i]);
81 }
82 printf("\n");
83

```

Code X

```

PS C:\Users\avik2\OneDrive\Documents\Desktop (2)\Code Playground\os assign;ments>c
d "c:\Users\avik2\OneDrive\Documents\Desktop (2)\Code Playground\os assign;ments\"
; if ($?) { gcc BankerAlgorithm.c -o BankerAlgorithm } ; if ($?) { .\BankerAlgori
thm }
Enter number of processes: 2
Enter number of resources: 2

Enter Allocation Matrix:
1
2
4
2

Enter Max Matrix:
4
2
5
3

Enter Available Resources:
5
0

System is in SAFE state
Safe sequence is:
P0 P1
PS C:\Users\avik2\OneDrive\Documents\Desktop (2)\Code Playground\os assign;ments>

```

C FIFO.c

Code

Run Debug Add View Lock ...

```
C FIFO.c > main()
1  #include <stdio.h>
2
3  int main() {
4      int pages[50], frames[10];
5      int n, f, i, j, k, pageFaults = 0;
6      int index = 0;
7
8      printf("Enter number of pages: ");
9      scanf("%d", &n);
10
11     printf("Enter the reference string:\n");
12     for(i = 0; i < n; i++) {
13         scanf("%d", &pages[i]);
14     }
15
16     printf("Enter number of frames: ");
17     scanf("%d", &f);
18
19     for(i = 0; i < f; i++) {
20         frames[i] = -1;
21     }
22
23     printf("\nPage Replacement Process:\n");
24
25     for(i = 0; i < n; i++) {
26         int found = 0;
27
28         for(j = 0; j < f; j++) {
29             if(frames[j] == pages[i]) {
30                 found = 1;
31                 break;
32             }
33         }
34
35         if(found == 0) {
```

```
PS C:\Users\avik2\OneDrive\Documents\Desktop (2)\Code Playground\os assign;ments>cd "c:\Users\avik2\OneDrive\Documents\Desktop (2)\Code Playground\os assign;ments\"
; if ($?) { gcc FIFO.c -o FIFO } ; if ($?) { .\FIFO }
Enter number of pages: 2
Enter the reference string:
2 3
Enter number of frames: 4

Page Replacement Process:
After page 2: 2 - - -
After page 3: 2 3 - -

Total Page Faults = 2
PS C:\Users\avik2\OneDrive\Documents\Desktop (2)\Code Playground\os assign;ments>
```

```

PS C:\Users\avik2\OneDrive\Documents\Desktop (2)\Code Playground\os assignments>
cd "c:\Users\avik2\OneDrive\Documents\Desktop (2)\Code Playground\os assignments\
" ; if ($?) { gcc Lru.c -o Lru } ; if ($?) { .\Lru }
Enter number of pages: 4
Enter page reference string: 2 3 6 5
Enter number of frames: 3
Frames: 2 - -
Frames: 2 3 -
Frames: 2 3 6
Frames: 5 3 6
Total Page Faults = 4
cd "c:\Users\avik2\OneDrive\Documents\Desktop (2)\Code Playground\os assignments\
" ; if ($?) { gcc Lru.c -o Lru } ; if ($?) { .\Lru }
Enter number of pages: 12
Enter page reference string:
2 4 6 3 5 7 3 9 6 4 1 7 4
Enter number of frames: Frames: 2 - - -
Frames: 2 4 - -
Frames: 2 4 6 -
Frames: 2 4 6 3
Frames: 5 4 6 3
Frames: 5 7 6 3
Frames: 5 7 9 3
Frames: 6 7 9 3
Frames: 6 4 9 3
Frames: 6 4 9 1
Frames: 6 4 7 1
Total Page Faults = 11
PS C:\Users\avik2\OneDrive\Documents\Desktop (2)\Code Playground\os assignments>

```

```
PS C:\Users\avik2\OneDrive\Documents\Desktop (2)\Code Playground\os assignments>
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